	CHAIN OF CUSTODY FORM (Original Stays With Samples)			US EPA Experimental Stream Facility 1003 US Highway 50, Milford, Ohio 45150 Phone # 513-965-4805 FAX # 513-965-4806					Originator:	
									Ship Date: Jan. 16, 18	Contact:
									Destination: Cincinnati	Phone #:
									Recipient: Sam Smith	E-Mail:
									Phone #: 513-569-7681	

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	CrossID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Preservative	Analyses Required
20180111	1640:00					DI	FIELDBLK		TNO2, TNO2-3, TNH4, TRP, TP, TN
20180113	09:50:00	FL	02	0.1	1	SW	UNK		"
20180113	10:26:00	FL	05	0.1	1	SW	UNK		"
20180113	11:05:00	FL	01	0.1	1	SW	UNK		"
20180113	11:34:00	FL	07	0.1	1	SW	UNK		"
20180113	12:07:00	FL	15	0.1	1	SW	UNK		"
20180113	12:25:00	FL	14	0.1	1	SW	UNK		"
20180113	12:46:00	FL	03	0.1	1	SW	UNK		"
20180113	13:15:00	FL	04	0.1	1	SW	UNK		"
20180113	13:47:00	FL	45	0.1	1	SW	UNK		"
20180113	14:17:00	FL	49	0.1	1	SW	UNK		"
20180113	15:22:00	FL	43	0.1	1	SW	UNK		"
20180113	17:08:00	FL	68	0.1	1	SW	UNK		"
20180116	12:29:00					DI	HOUSEBLK		"
20180111	16:40:00					DI	FIELDBLK	2MHCl	TOC
20180113	09:50:00	FL	02	0.1	1	SW		2MHCl	TOC
20180113	10:26:00	FL	05	0.1	1	SW		2MHCl	TOC
20180113	11:05:00	FL	01	0.1	1	SW		2MHCl	TOC
20180113	11:34:00	FL	07	0.1	1	SW		2MHCl	TOC
20180113	12:07:00	FL	15	0.1	1	SW		2MHCl	TOC
20180113	12:25:00	FL	14	0.1	1	SW		2MHCl	TOC
20180113	12:46:00	FL	03	0.1	1	SW		2MHCl	TOC
20180113	13:15:00	FL	04	0.1	1	SW		2MHCl	TOC
20180113	13:47:00	FL	45	0.1	1	SW		2MHCl	TOC
20180113	14:17:00	FL	49	0.1	1	SW		2MHCl	TOC
20180113	15:22:00	FL	43	0.1	1	SW		2MHCl	TOC
20180113	17:08:00	FL	68	0.1	1	SW		2MHCl	TOC
20180116	12:29:00					DI	HOUSEBLK	2MHCl	TOC

	Relinquished By:	Date:	Time:	Received By:	Date:		Time:

CHAIN OF CUSTODY FORM
(Original Stays With Samples)

US EPA Experimental Stream Facility
1003 US Highway 50, Milford, Ohio 45150
Phone # 513-965-4805
FAX # 513-965-4806

Originator:			
Ship Date:	January 16th	Contact:	
Destination:	Cincinnati	Phone #:	
Recipient:	Sam Smith	E-Mail:	
Phone #:	513-569-768		

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	Cross ID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Analyses Required	*Volume of liquid sample filtered (mL)	
20180111	16:40:00					DI	FIELD BLK	Chlorophyll	331.49	
20180113	09:50:00	FL	02	0.1	1	SW	UNK	"	306.52	
20180113	10:26:00	FL	05	0.1	1	SW	UNK	"	325.94	
20180113	11:05:00	FL	01	0.1	1	SW	UNK	"	308.83	
20180113	11:34:00	FL	07	0.1	1	SW	UNK	"	251.48	
20180113	12:07:00	FL	15	0.1	1	SW	UNK	"	266.95	
20180113	12:25:00	FL	14	0.1	1	SW	UNK	"	269.7	
20180113	12:46:00	FL	03	0.1	1	SW	UNK	"	295.75	
20180113	13:15:00	FL	04	0.1	1	SW	UNK	"	280.64	
20180113	13:47:00	FL	45	0.1	1	SW	UNK	"	218.44	
20180113	15:22:00	FL	43	0.1	1	SW	UNK	"	175.91	
20180113	14:17:00	FL	49	0.1	1	SW	UNK	"	165.72	
20180113	17:08:00	FL	68	0.1	1	SW	UNK	"	174.6	

Relinquished By:	Date:	Time:	Received By:

This Sampling occurred 1st day of thaw. Lake Previously Frozen for Several Days.

01/12/2018

2? 14:10 25 mL

GAS TRAP DATA SHEET

Lake:			1/12/18	1/13/18		
Site	**Lat**	**Long**	**Deployment time/date**	**Retrieval time/date**	**Volume recovered (mL)**	**Exetainer codes**
24	35.942100	78.588103	10:34	09:15	25	24.BT.1
16	35.956696	78.594491	10:48	09:31	25 25 14	16.BT.1 16.BT.2
02	35.976198	78.588983	10:59	09:50	0	TRAP broken - TRAP
05	35.97017	78.62231	11:12	10:26	25 20	05.BT.1 05.BT.2
12	35.965545	78.633990	11:21	10:37	10	
13	36.003656	78.646320	11:33	10:54	0	
01	36.002155	78.63452	11:44	11:05	0	
21	36.008900	78.628055	11:52	11:15	1	
07	36.005660	78.667918	12:05	11:34	14	
11	36.019453	78.71710	12:34	11:55	0	1 Leg off trap
15	36.023012	78.7146608	12:42	12:07	0	
14	36.040123	78.717818	12:55	12:25	25 4	14.BT.1
03	36.032271	78.736070	13:28	12:46	0	
08	36.019051	78.748751	13:39	13:03	2	
04	36.013611	78.759898	13:51	13:15	2	
10	36.040293	78.755950	13:11	13:35	13	
45	36.044158	78.758283	14:17	13:47	0	
57	36.048440	78.752226	14:28	14:00	2	1 leg off trap.
49	36.051346	78.757018	14:37	14:17	8	Trap Moved
55	36.053225	78.761636	14:42	14:38	8	
61	36.053536	78.759471	14.47	14:10	25	61.BT.1
63	36.055140	78.770235	15:01	15:09	3	Trap Moved Visible BPs?
54	36.058018	78.765255	15:13	14:53	25 25 10	54.BT.1 54.BT.2
43	36.061634	78.773188	15:28	15:22	25 25 25 27	43.BT.1 43.BT.2 43.BT.3
58	36.066680	78.775853	15:46	16:05	25 25 25 21	58.BT.1 58.BT.2 58.BT.3
47	36.068645	78.769615	16:02	14:21	14	
68	36.065373	78.766938	16:09	17:08	16	
56	36.072510	78.7662551	16:15	16:47	25 25 7	56.BT.1 56.BT.2
52	36.079053	78.761305	16:29	16:33	25 25 25 135	52.BT.1 52.BT.2 52.BT.3
59	36.062861	78.771298	16:58	15:35	25 25 25 +?	59.BT.1 59.BT.2 59.BT.3

Trap in bottom

TRAP Moved Air from M

DISSOLVED GAS AND AIR DATA SHEET

Lake:	FALS				Date:	9/13/18		
¥Site	Dissolved gas Exetainer codes		Water Vol/ Helium Vol	Water Temp (degC)	Atmospheric Air Exetainer Codes			BP (mm Hg)
2	02.DG.1	02.DG.2	115/25	7.5	02.AA.1	02.AA.2		
5	05.DG.1	05.DG.2	"	6.9				
1	01.DG.1	01.DG.2	"	7.3				
7	07.DG.1	07.DG.2	"	7.0				
15	15.DG.1	15.DG.2	"	6.1				
14	14.DG.1	14.DG.2	"	8.5				
3	03.DG.1	03.DG.2	"	8.0				
4	04.DG.1	04.DG.2	"	8.3				
45	45.DG.1	45.DG.2	"	8.0				
49	49.DG.1	49.DG.2	"	7.4				
43	43.DG.1	43.DG.2	"	6.2	43.AA.1	43.AA.2		
47	47.DG.1	47.DG.2	"	7.1				

¥Site: all dissolved gas samples collected at 0.1m depth
Boat motor should be off for 2 minutes before sample collection
Goal is to inject 20mL of gas into exetainer

Signature Date

GREENHOUSE GAS ANALYZER AND GPS DATA SHEET

FILE NAMES
Lake: FALLS
Sample Date: 8/13/18
GGA file 999-2018-01-13-f0000.txt
gps file name

GPS Time 14:17:00
LGR Time 09:15:32

DEPLOYMENT DATA

Sample site	Chamber start time	Bubbling observed (Y/N)	GPS logger started	Chamber volume (L)	CH_4 and CO_2 profile checked	WATER T. (°C)
24	09:16:53	N	✓	45	✓	5.3
16	09:31:00	✓	✓	~~45~~ 50	✓	6.0
02	09:50:48	N	✓	50	✓	8.3
05	10:26:22	N	✓	50	✓	7.6
12	10:37:45	N	✓	50	✓	5.8
13	10:54:25	N	✓	50	✓	6.5
01	11:05:45	N	✓	50	✓	6.0
21	11:16:45	N	✓	50	✓	6.8
07	11:34:45	N	✓	50	✓	5.6
11	11:56:00	N	✓	50	✓	6.0
15	12:07:45	N	✓	50	✓	6.0
14	12:25:00	N	✓	50	✓	~~6.0~~ 9.9
03	12:47:00	N	✓	45	✓	8.0
08	13:03:45	N	✓	45	✓	7.2
04	13:15:45	N	✓	50	✓	8.6
10	13:35:30	N	✓	50	✓	7.3
45	13:47:10	N	✓	50	✓	7.6
57	14:01:00	N	✓	50	✓	8.7
49	14:17:45	N	✓	50	✓	8.7
61	14:28:15	N	✓	50	✓	8.5
55	14:38:10	N	✓	50	✓	8.5
54	14:53:00	N	✓	50	✓	8.0
63	15:09:00	N	✓	50	✓	8.5
43	15:22:00	N	✓	50	✓	8.5
59	15:35:30	N	✓	50	✓	9.1
58	16:05:30	N	✓	50	✓	10.1
47	16:21:00	N	✓	50	✓	9.6
52	16:33:09	N	✓	50	✓	10.5
56	16:47:00	N	✓	50	✓	9.8
68	NO SAMPLE					

↳ Signal decrease @ end of sample. Connection (VNC) failure @ restart. Low Power.

SONDE DATA SHEET

Lake: Sample Date: 01/12/2017

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond (µs cm^{-1})	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
24	1.5	0.1	4.82	10.2	79.6	0.085	7.47	5.4	3.2	
16	6.1	"	7.83	10.21	85.6	0.086	7.83	5.0	2.1	
02	3.1	"	7.0	11.57	96.3	0.089	7.71	3.1	2.5	
05	3.2	"	7.63	9.96	83.5	0.087	7.88	5.1	1.9	
12	5.2	"	6.49	10.55	86.8	0.088	7.78	5.2	1.8	
13	5.6	"	8.23	12.78	108.4	0.090	7.71	10.5	2.1	
01	3.9	"	5.45	11.87	94.5	0.091	7.72	10.1	2.2	
21	4.4	"	5.79	11.51	95.2	0.092	7.76	6.1	2.4	
07	9.0	20 ft	4.40	11.57	89.2	0.100	7.66	19.2	131.1	
07		17 ft	4.42	12.25	94.5	0.100	7.64	18.6	2.7	
07		14 ft	4.51	12.36	95.6	0.100	7.63	18.5	2.7	
07		11 ft	4.51	12.36	95.6	0.100	7.63	18.9	2.8	
07		8 ft	4.56	12.39	96.0	0.100	7.63	19.6	2.9	
07		5 ft	4.66	12.4	96.5	0.100	7.63	18.5	2.7	
07		2.5 ft	4.80	12.37	96.4	0.100	7.63	12.0	2.7	
07		0.5 ft	4.82	12.60	98.3	0.100	7.63	6.4	2.7	
11	2.6	0.1 m	4.73	11.80	93.8	0.117	7.51	5.3	2.2	
15	3.8	"	4.67	12.54	97.9	0.117	7.73	4.4	2.6	
14	1.0	"	7.86	12.87	108.4	0.109	7.66	0.065	5.0	
03	0.8	"	4.88	12.58	98.7	0.108	7.74	4.3	4.8	
08	3.4	"	5.29	13.47	106.6	0.148	7.71	5.7	4.0	
04	0.8	"	6.30	13.80	113.5	0.144	7.80	7.0	6.3	
10	1.2	"	6.0	12.45	100.7	0.175	7.98	9.8	5.7	
45	1.6	"	6.46	12.85	104.8	0.187	7.91	6.8	6.7	
57	1.3	"	7.26	12.48	104.0	0.198	7.96	10.5	8.4	
49	2.0	"	6.69	12.66	103.6	0.235	7.91	15.5	11.4	
55	1.4	"	7.41	13.59	113.9	0.240	7.90	10.0	14.8	

SONDE DATA SHEET

Lake: Sample Date: 01/12/2017 ← 2018 ??

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond (µs cm^{-1})	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
61	1.4	0.1	7.36	13.58	113.2	0.242	7.94	8.0	12.6	
63	1.5	"	7.10	13.50	115.0	0.244	7.98	5.5	15.4	
54	0.7	"	7.38	12.84	107.1	0.243	8.13	7.0	17.0	
43	0.7	"	8.07	12.45	106.3	0.256	8.08	11.6	23.3	
58	0.8	"	8.86	13.54	117.0	0.298	8.08	15.7	24.6	
47	1.2	"	10.35	13.05	116.8	0.256	8.14	11.5	25.2	
68	1.5	"	8.94	12.66	109.8	0.245	8.17	9.5	18.0	
56	1.5	"	8.78	13.31	114.7	0.231	8.12	8.4	12.4	
52	0.9	"	11.15	13.35	121.8	0.237	8.10	11.2	18.9	
59	0.8	"	9.15	14.01	121.8	0.397	8.20	12.2	24.1	

Date 12/8/17

Cal cylinder serial # CC163816

Dilution system serial # A20808

LGR file JPA-2017-12-18_f0000.txt

Screenshots of Laser Alignment.

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	-0.01	-0.01	9:12	9:33
9	7.0	2.4	23.7	2.36	23.3	9:50	9:52
8.5	7.0	2.5	25.1	2.45	24.4	~~9:52~~ 10:25	~~9:58~~
8	7.0	2.7	26.7	2.61	26.1	10:20	10:24 10:29
7.5	7.0	2.8	28.4	2.8	28.0	10:16	10:20
7	7.0	3.0	30.5	3.00	29.8	10:12	10:16
6.5	7.0	3.3	32.8	3.24	32.3	10:07	10:11
6	7.0	3.6	35.5	3.52	35.0	10:03	10:07
~~5.5~~	~~7.0~~	~~3.9~~	~~38.8~~				
5	7.0	4.3	42.7	4.22	42.0	09:59	10:03
53	7.0	4.0	40.2	4.00	39.70	9:52	9:58

Screenshot of Alignment

Signature _____

Date 1/16/18
Cal cylinder serial # CC163816
Dilution system serial # EPA A20808
LGR file 259 - 2018-01-16_f0000.txt

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	-0.01	0.5	15:10	15:16:15
9	7.0	2.4	23.7	2.32	22.9	15:17:15	15:21:00
8.5	7.0	2.5	25.1	2.53	25.1	15:21:00	15:23:50
8	7.0	2.7	26.7	2.70	26.8	15:23:50	15:27:15
7.5	7.0	2.8	28.4	2.88	28.7	15:27:15	15:29:48
7	7.0	3.0	30.5	3.08	30.6	15:29:48	15:32:00
6.5	7.0	3.3	32.8	3.30	32.8	15:32:00	15:34:33
6	7.0	3.6	35.5	3.53	35.3	15:34:00	15:38:45
5.5	7.0	3.9	38.8	3.84	38.2	15:38:45	15:40:20
5	7.0	4.3	42.7	4.21	42.0	15:40:20	15:42:57

Signature _____

Pre deployment

Temp (TC) Temp (probe)
20.60 20.56

ATTACHMENT A
YSI 6600 SONDE CALIBRATION LOG SHEET

SONDE ID: STD 16 **DATE:** 12/18/12 **TECHNICIAN:** JTW

DATE / TIME OK? (YES) / NO

BATTERY VOLTS: 12.5 (Change if < 10.0 v) **CHANGED BATTERIES?** YES / (NO)

TURBIDITY WIPER: PARKS 180° FROM OPTICS? (YES) / NO **CHANGED?** YES / (NO)

CHLOROPHYLL WIPER: PARKS 180° FROM OPTICS? (YES) / NO **CHANGED?** YES / (NO)

DISSOLVED OXYGEN: "DO Sat %" **BAROMETRIC PRESSURE:** 755.7 mm Hg

SENSOR OUTPUT TEST: PASS / FAIL (For Rapid-Pulse Probes, change membrane if failed)

Record first 10 readings from Discrete Sampling:

_____ _____ _____ _____ _____

_____ _____ _____ _____ _____

DO MEMBRANE CHANGED? YES / (NO) **PROBE RE-SURFACED?** YES / (NO) **DATE/TIME:** _____

[~97] **INITIAL:** 98.0 **CALIBRATED:** 99.4 **POST-DEPLOYMENT:** _____
DO CHARGE [50 ± 25]: _____ **DO GAIN** [1.0 ± 0.35]: 1.04602

DEPTH: "Pressure" [m] **PRESSURE OFFSET** [VENTED = 0 ± 6 / NON-VENTED = -14.7 ± 6]: 0.2323
[0] **INITIAL:** -0.546 **CALIBRATED:** 0.00 **POST-DEPLOYMENT:** _____

CONDUCTIVITY: "SpCond" [mS/cm] **CONDUCTIVITY CELL CONSTANT** [5.0 ± 0.45]: 4.8851 ~~4.82544~~ Temp 20.3 initial

[10] **INITIAL:** 1387 **CALIBRATED:** 1.413 **POST-DEPLOYMENT:** _____
standard = 1.413 @ 25°C

TURBIDITY: "Turb+NTU" [NTU]

[0] **INITIAL:** 0 **CALIBRATED:** 0 **POST-DEPLOYMENT:** _____
[~~1000~~ 126] **INITIAL:** 125.5 **CALIBRATED:** 126 **POST-DEPLOYMENT:** _____

CHLOROPHYLL: "Chlor ug/L" [ug/L]

[0] **INITIAL:** -0.1 **CALIBRATED:** X **POST-DEPLOYMENT:** _____
[113] **INITIAL:** _____ **CALIBRATED:** _____ **POST-DEPLOYMENT:** _____

---

pH: "pH"

[7] **INITIAL:** 6.928 **CALIBRATED:** 7.0 **POST-DEPLOYMENT:** _____ **mV** [0 ± 40]: -35.2
[10] **INITIAL:** 9.99 **CALIBRATED:** _____ **POST-DEPLOYMENT:** _____ **mV** [-177 ± 40]: -202.3

ORP: "ORP mV" **ORP mV OFFSET [0 ± 100]:** _____

Printed copies of this document are uncontrolled. All users are responsible for confirming version status against the electronic version in the document control system.

1/12/18

~~12/13/2017~~

	# Samples collected			
Site#	Bubble Trap	Dissolved Gas	Water	Air
52	3			
56	3			
68	3			
47	3	2	1*	
58	3			
59	3			
43	3	2	1	5
54	3			
63	3			
61	3			
55	3			
49	3	2	1	
57	3			
45	3	2	1	
10	3			
4	3	2	1	
8	3			
3	3	2	1	
14	3	2	1	
15	3	2	1	
11	3			
7	3	2	1	
21	3			
1	3	2	1	
13	3			
12	3			
5	3	2	1	
2	3	2	1*	5
16	3			
24	3			

*Collect extra water for sonde calibration.